

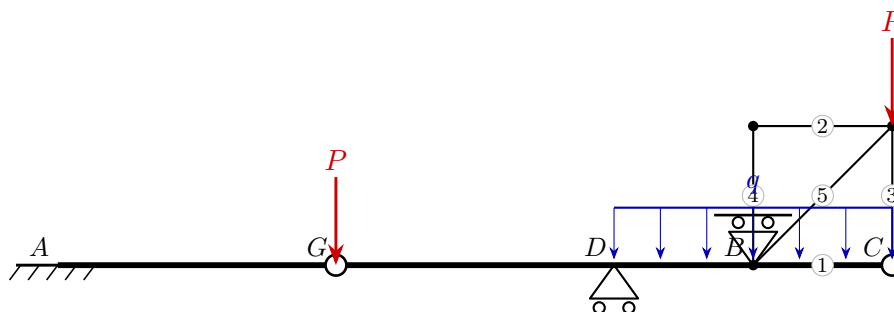
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 3 - (5 + 4) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

 $A: R_x = 0, R_y = -15, M = -45 \quad D: R_x = 0, R_y = 52.5 \quad B: R_x = 0, R_y = 0$
 $C: H_x = 0, H_y = -15 \quad G: H_x = 0, H_y = 22.5$

3. Truss member forces

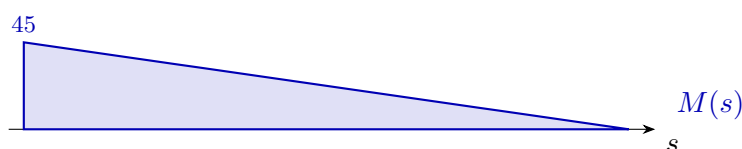
Member	Force S [kN]	Type
1	0	zero
2	0	zero
3	-15	compression
4	0	zero
5	0	zero

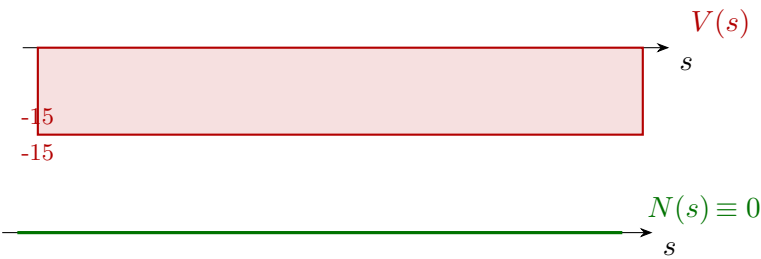
red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

Beam 1





Beam 2

